**Topic**: Separable equations

**Question**: Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan^2 y}$$

**Answer choices**:

A $\tan y - y = C \tan x$

B $\tan y - y = \tan x + C$

C $y = \tan x + C$

D $y(\tan y - 1) = \tan x + C$

© Krista King Math

**Solution**: B

Start by separating variables, putting $y$ terms on the left and $x$ terms on the right.

$$\frac{dy}{dx} = \frac{\sec^2 x}{\tan^2 y}$$

$$dy = \frac{\sec^2 x}{\tan^2 y}\, dx$$

$$\tan^2 y \; dy = \sec^2 x \; dx$$

Now integrate both sides, adding the constant of integration $C$ to the right side.

$$\int \tan^2 y \; dy = \int \sec^2 x \; dx$$

$$\int \sec^2 y - 1 \; dy = \int \sec^2 x \; dx$$

$$\tan y - y = \tan x + C$$

We'll let this be the implicit solution to the separable differential equation.

© Krista King Math

**Topic**: Separable equations

**Question**: Find the solution to the separable differential equation.

$$\frac{du}{dv} = \frac{3v\sqrt{1+u^2}}{u}$$

**Answer choices**:

A $u = \sqrt{\frac{3}{2}v^2 + C}$

B $u = \sqrt{\frac{3}{2}v^2 + C}$

C $u = \sqrt{\left(\frac{3}{2}v^2 + C\right)^2 - \frac{3}{2}}$

D $u = \sqrt{\left(\frac{3}{2}v^2 + C\right)^2 - 1}$

© Krista King Math

**Solution**: D

Start by separating variables, putting $u$ terms on the left and $v$ terms on the right.

$$\frac{du}{dv} = \frac{3v\sqrt{1+u^2}}{u}$$

$$du = \frac{3v\sqrt{1+u^2}}{u}\, dv$$

$$u\ du = 3v\sqrt{1+u^2}\ dv$$

$$\frac{u}{\sqrt{1+u^2}}\ du = 3v\ dv$$

Now integrate both sides, using a substitution for the left side with $k = 1 + u^2$ and $du = dk/2u$, and adding the constant of integration $C$ to the right side.

$$\int \frac{u}{\sqrt{k}}\left(\frac{dk}{2u}\right) = \int 3v\ dv$$

$$\frac{1}{2}\int k^{-\frac{1}{2}}\ dk = \int 3v\ dv$$

$$k^{\frac{1}{2}} = \frac{3}{2}v^2 + C$$

Back-substitute for $k = 1 + u^2$.

$$(1+u^2)^{\frac{1}{2}} = \frac{3}{2}v^2 + C$$

© Krista King Math

$$\sqrt{1+u^2} = \frac{3}{2}v^2 + C$$

Solve for $u$ to get the explicit the solution to the separable differential equation.

$$1+u^2 = \left(\frac{3}{2}v^2 + C\right)^2$$

$$u^2 = \left(\frac{3}{2}v^2 + C\right)^2 - 1$$

$$u = \sqrt{\left(\frac{3}{2}v^2 + C\right)^2 - 1}$$

© Krista King Math

**Topic**: Separable equations

**Question**: Which first order differential equation is not separable?

**Answer choices**:

A $y' = e^{x-y}$

B $y' = 1 + \frac{x}{y}$

C $y' = y + \frac{y}{x}$

D $y' = \frac{x}{1+y}$

© Krista King Math

**Solution**: B

Answer choice A can be rewritten as

$$y' = e^{x-y}$$

$$\frac{dy}{dx} = e^x e^{-y}$$

$$\frac{dy}{dx} = \frac{e^x}{e^y}$$

$$e^y \, dy = e^x \, dx$$

The variables in answer choice B can't be separated, so the equation isn't separable.

Answer choice C can be rewritten as

$$y' = y + \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{xy}{x} + \frac{y}{x}$$

$$\frac{dy}{dx} = \frac{y(x+1)}{x}$$

$$\frac{1}{y} \, dy = \frac{x+1}{x} \, dx$$

Answer choice D can be rewritten as

$$y' = \frac{x}{1+y}$$

© Krista King Math

$$dy = \frac{x}{1+y}\,dx$$

$$(1+y)\,dy = x\,dx$$

© Krista King Math